

Strategic Fee-Level Choice under Incomplete Information: An Anti-Coordination Game on Algorand

Lin Zhang

Duke Kunshan University
Kunshan, China

lin.zhang@dukekunshan.edu.cn

Zijie Wang

Duke Kunshan University
Kunshan, China

zijie.wang@dukekunshan.edu.cn

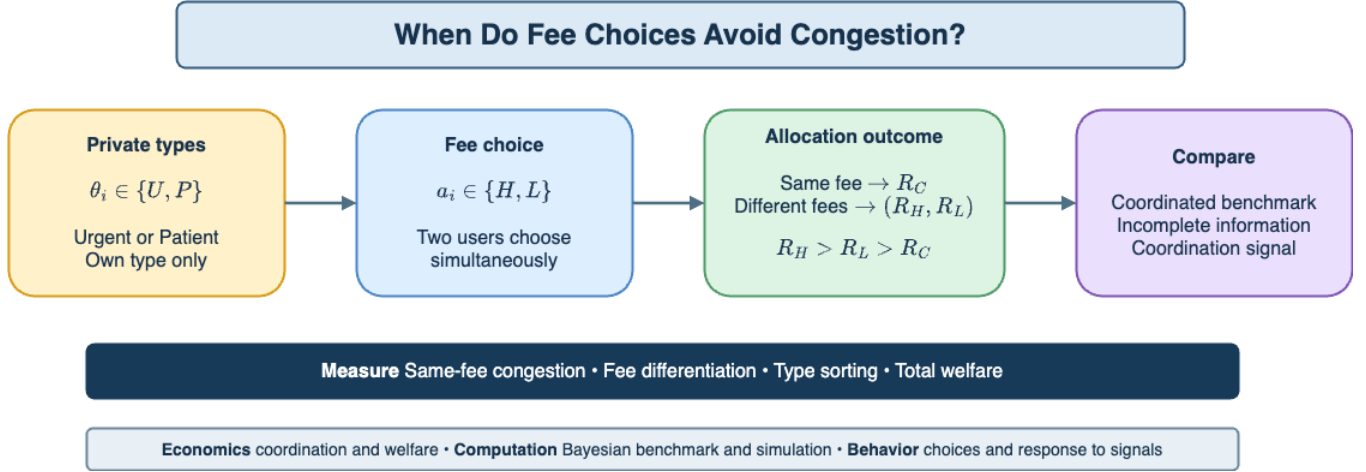


Figure 1: Research design. Two users with private Urgent or Patient types simultaneously choose High or Low fees. We compare a coordinated benchmark, decentralized choice under incomplete information, and a coordination signal using congestion, fee differentiation, type sorting, and welfare.

Submission to: COMSCI/ECON 206 — Computational Microeconomics.

Course session: Autumn 2026 Session 1. **Team:** Team7. **Symposium:** September 28 Symposium.

Course instructor: Professor Luyao Zhang. **Assignment:** PS2 collaborative research proposal.

1 Three connected research questions and verified gap

Blockchain transaction capacity is limited, so users may compete for transaction priority through their fee choices [10]. We study a simplified version of this problem in the Algorand setting [6]. Two users simultaneously choose a *High* (H) or *Low* (L) fee. Each user privately observes whether the transaction is Urgent or Patient, representing different costs of delay. When both users choose the same fee level, they receive a congestion payoff; when they choose different levels, transaction priority is differentiated.

Our shared question is how private transaction urgency affects decentralized priority allocation and whether a coordination signal can improve it. **Q1—Economics:** How do urgency and value affect coordination and welfare? **Q2—Computation:** What equilibrium behavior follows from the payoff structure and type beliefs, and how do different information rules affect congestion and welfare? **Q3—Behavior:** Do people follow the type-dependent rational

benchmark, and can a coordination signal reduce coordination failures?

Existing work provides foundations for Bayesian games, coordination, and transaction-fee mechanism design [1, 7, 10]. We focus on a narrower question: whether privately informed users can successfully differentiate simple High and Low fee choices when matching choices create congestion. The three questions connect the theoretical benchmark, its computational implications, and possible departures in human decisions.

2 Answer to Q1: incentives and strategic model

There are two users $i \in \{1, 2\}$. Each privately observes a type $\theta_i \in \{U, P\}$, where U is Urgent and P is Patient, and simultaneously chooses a fee level $a_i \in \{H, L\}$. Conditional on types, the payoff matrix is

	H	L
H	(R_C, R_C)	(R_H, R_L)
L	(R_L, R_H)	(R_C, R_C)

where R_C is the payoff from a same-fee outcome and

$$R_H > R_L > R_C.$$

A user therefore prefers the High position when the other user chooses Low, but prefers Low when the other user chooses High. Under complete information, (H, L) and (L, H) are pure Nash equilibria [9]. The game is an anti-coordination problem because both users benefit from choosing different fee levels, while the remaining

problem is who takes each role. The symmetric mixed equilibrium chooses H with

$$p^* = \frac{R_H - R_C}{R_H + R_L - 2R_C}.$$

With private types, each user knows the value of priority for their own transaction but not the other user's type. The relevant solution concept is therefore Bayesian Nash equilibrium [7]. Type-specific payoffs allow an Urgent user to value H more strongly than a Patient user, so equilibrium strategies can map different types to different fee choices.

The three theoretical lenses evaluate different parts of this problem. Game theory gives the best-response and Bayesian equilibrium benchmarks. Social choice asks whether decentralized choices produce the welfare-maximizing allocation, especially whether higher urgency is matched with higher priority. Mechanism design then considers whether a nonbinding coordination signal can reduce same-fee outcomes without directly revealing private types [2, 3].

3 Answer to Q2: computation, allocation comparison, and artifacts

The computational notebook implements the same one-shot Bayesian game with type-specific payoffs. The baseline calibration uses

$$(R_H, R_L, R_C)_U = (8, 2, 1), \quad (R_H, R_L, R_C)_P = (5, 4, 1),$$

so the Urgent type gains more from receiving the High position. If q is the believed probability that the other user chooses H , the expected payoffs for type θ are

$$\begin{aligned} EU(H \mid \theta, q) &= qR_C(\theta) + (1 - q)R_H(\theta), \\ EU(L \mid \theta, q) &= qR_L(\theta) + (1 - q)R_C(\theta). \end{aligned}$$

The notebook uses these expected payoffs to check type-dependent best responses and pure Bayesian equilibria. Under the baseline parameters, one pure Bayesian equilibrium is

$$U \rightarrow H, \quad P \rightarrow L.$$

The Urgent-type prior is then varied to test how strongly this result depends on the assumed population mix.

We compare three allocation environments. A coordinated complete-information case provides a benchmark in which types are known and an efficient H/L allocation can be selected. Under incomplete information, users choose according to the Bayesian type strategy. The signal condition keeps types private but gives differentiated recommendations that users follow with a specified probability. Each simulated pair is independent, so the simulation does not turn the model into a repeated strategic game.

The scarce resource is transaction priority. We compare same-fee congestion, successful H/L differentiation, type-priority sorting, individual payoff, and total welfare. A planner benchmark also selects the differentiated assignment with the larger total payoff, allowing us to measure how far each treatment is from the welfare-maximizing allocation. Sensitivity checks vary the type prior and signal compliance. These outputs are conditional predictions from the model, not participant data or evidence about current Algorand network performance.

Computational scientist—GitHub: https://github.com/z2005lin/PS2_Zijie_Lin

Runnable notebook: <https://colab.research.google.com/drive/1O>

x65uxV-g7nPSgpChWZFGmYOfYQTQoCi?usp=sharing

Behavioral scientist—Hugging Face:

Interdisciplinary scholar—A0 poster:

4 Answer to Q3: behavior and interdisciplinary integration

The rational benchmark assumes that users form beliefs about the other user's choice and select the action with higher expected payoff. Human decisions may instead reflect focal choices, uncertainty, limited strategic reasoning, or learning [4, 12].

The behavioral study records each fee choice together with the user's belief that the other user chooses H , confidence, and a short decision reason. These measures help distinguish Bayesian best responses from simpler decision rules, since the same H/L choice can result from different reasoning. The signal condition then tests whether recommendations improve coordination when types remain private. Human behavioral effects remain hypotheses until participant data are collected.

5 Advanced development, real-world impact, and roadmap

This project connects a Bayesian fee-choice model with computational and behavioral tests of priority allocation. If the mechanism remains useful after human testing and better calibration, it may help study how wallet interfaces present fee choices or coordination recommendations. The main risks are unrealistic payoff calibration, unequal access to priority, and the gap between a two-user model and a live blockchain network. Sensitivity analysis and behavioral testing provide ways to identify these limits.

The next step is to compare human choices with the Bayesian benchmark and revise the signal or payoff assumptions when needed. Later work can extend the model to more users and richer fee choices.

Expected 2056 Nobel Prize and Turing Award contribution statement. We hope to develop transparent and testable tools for allocating scarce digital resources under private information.

Open Science Statement

GitHub and the notebook document the model, parameters, code, and computational outputs. Hugging Face provides the behavioral interface, and the A0 poster summarizes the combined evidence.

Statement of Contribution to the UN's SDGs

The project relates to SDG 9 through reliable digital infrastructure, with access effects requiring further evaluation.

Author Notes

Acknowledgements

Course instructor: Professor Luyao Zhang. We acknowledge OpenAI Codex for assistance on September 25–26, 2026 with transferring the authors’ proposal into the required LaTeX structure, refining the teaser figure, and checking layout. No empirical result was generated or verified by this assistance. No peer, discussant, community, or industry feedback has yet been received. The authors remain responsible for every claim, citation, computation, and remaining error.

Team and individual contributions

Team 7 jointly narrowed the study to a two-user High/Low fee-choice game and specified the equilibrium and experiment plan. **Lin Zhang** is responsible for the future GitHub implementation, runnable notebook/Colab, and Hugging Face behavioral artifact. **Zijie Wang** developed the proposal and Overleaf submission, coordinated the model framing and teaser, and will produce the A0 poster. Both authors will calibrate the game, verify outputs, interpret results, and revise claims after feedback.

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A Technical details and reproducibility

For type θ , let $R_H(\theta)$ denote the reward from being the high-fee user in a differentiated pair, $R_L(\theta)$ the reward from being the low-fee user, and $R_C(\theta)$ the reward under same-fee congestion. The benchmark condition $R_H > R_L > R_C$ produces two asymmetric pure equilibria and the mixed probability shown in Section 2. A Bayesian

solution additionally requires the type prior $\mu = \Pr(\theta = U)$ and all type-contingent rewards.

The preregistered implementation will: (1) specify the H/L mapping and payoff table; (2) solve complete-information equilibria; (3) compute type-dependent Bayesian best responses; (4) randomize participants into the three information treatments; (5) record choices, beliefs, payoffs, matching, and round order; and (6) reproduce all tables from a fresh run. Missing inputs are the Algorand fee mapping, congestion and confirmation rule, payoff calibration, type prior, sample size, seeds, and ethics/power-analysis decisions. No code, log, parameter file, seed, or fresh-run output currently exists.

A.1 AI-use disclosure

OpenAI Codex was used to restructure the supplied proposal, formalize the clarified payoff matrix, draft truthful status language, redraw the teaser, and inspect document layout. The authors instructed that no missing data, simulation, laboratory outcome, feedback, or link be invented. Human verification remains required for payoffs, citations, calibration, code, and final claims.

B Cumulative Intellectual Development

The earlier draft framed a multi-round Submit/Wait congestion game. On September 26, 2026, the authors clarified a different and narrower research object: in a single round, two users simultaneously choose High Fee or Low Fee; matching choices congest and yield low rewards, whereas different choices allow the high-fee user to receive a high reward. The current version therefore replaces the dynamic state transition with a static Bayesian anti-coordination matrix, adds the missing low-fee payoff symbol, and organizes the empirical plan around information treatments. This change is a model correction, not an empirical finding.

C Open-source and three-lens inspiration

The verified foundations include Nash and Bayesian equilibrium [7, 9], correlated recommendations and information design [2, 3], induced-value experimentation [11], laboratory coordination evidence [1, 5, 8], behavioral strategic choice [4, 12], Algorand’s protocol design [6], and blockchain fee-mechanism analysis [10]. **Game theory** tests mutual best responses; **social choice** evaluates type sorting and welfare; **mechanism design** evaluates coordination signals. Repository URLs, licenses, versions, and protocol-document versions will be recorded when implementation begins.

D Review response and revision record

No peer, discussant, or instructor review has been received as of September 26, 2026. The following are internal decisions only. After the symposium, every question and evaluation will be logged with its source, Keep/Revise/Investigate decision, changed location, and rerun check.

Table 1: Internal pre-review decisions.

Decision	Location	Reason and change
Keep	Sections 1–2	Retain private user types and simultaneous binary fee choice.
Revise	Throughout	Replace Submit/Wait dynamics with a static H/L anti-coordination game.
Investigate	Appendix A	Calibrate payoffs, type prior, fee mapping, sample size, and signal design.

E Structured abstract and cumulative proposal map

This map provides a self-contained account of the proposal and preserves the template’s eight-part cumulative structure. It separates established foundations, completed proposal work, testable expectations, and evidence that remains unavailable.

Table 2: Appendix E structured abstract template; retain the familiar PS1 structure.

Proposal element	Current statement / change
Background	Blockchain transaction capacity is scarce, so fee mechanisms determine which transactions receive priority. Existing work supplies Nash and Bayesian equilibrium, correlated recommendations, experimental coordination evidence, and formal transaction-fee design. The unresolved issue here is whether two privately informed users can reliably differentiate between High and Low fees when equal choices generate congestion.
Motivation and application	The application is a stylized Algorand fee-choice interface for an urgent and a patient user. A coordination failure occurs when both select the same fee level and receive the low congestion payoff. The model is deliberately narrower than a live protocol: its value is to isolate strategic uncertainty before adding more users, continuous fees, or network dynamics.
Three connected questions	Economics asks how private urgency changes equilibrium selection, efficient sorting, and welfare. Computation asks how to solve the pure, mixed, and type-dependent Bayesian equilibria and reproduce treatment predictions. Behavioral science asks whether focal choices, ambiguity, limited strategic reasoning, or learning cause systematic departures and whether a coordination signal reduces them.
Method and model assumptions	Two users simultaneously choose H or L after privately observing type U or P . The benchmark assumes $R_H > R_L > R_C$ and a common prior over types. We compare complete information, incomplete information, and incomplete information plus a signal, using random rematching, induced experimental rewards, belief elicitation, and preregistered analysis.
Results or expected results	Completed outputs are the symbolic payoff matrix, two asymmetric pure equilibria, the symmetric mixed-strategy expression, treatment design, teaser, and measurement plan. We expect incomplete information to increase same-fee congestion, heterogeneity to create partial sorting, and a signal to improve differentiation and welfare. These are falsifiable hypotheses; no numerical or participant result exists.
Intellectual merits	The project connects a tractable anti-coordination game to Bayesian information design and controlled experimental evidence. It makes the welfare criterion explicit: successful differentiation and assignment of H to the higher-urgency user. The contribution is not a claim about Algorand performance, but a reproducible bridge from equilibrium assumptions to observable choices and mechanism comparison.
Practical impacts	If externally validated, the findings could inform how wallets display discrete fee options or provide nonbinding recommendations. Relevant outcomes include congestion, type-specific access, user welfare, and robustness across payoff calibrations. Risks include oversimplifying the live network, embedding unfair priority, or treating laboratory coordination as field evidence; open parameters and versioned code provide correction routes.
Feedback received and revision made	No peer, instructor, or symposium feedback has yet been received. Internal review on September 26 replaced the earlier dynamic Submit/Wait model with the clarified one-shot High/Low anti-coordination game, added the low-fee payoff and equilibrium conditions, and redesigned the experiment around information treatments. Future external feedback will be logged as Keep, Revise, or Investigate.

F PS2 extension record: experiment, linked artifacts, poster, and symposium

F.1 Experimental record

No laboratory experiment has been run. There are no actual types, choices, bids, payments, utilities, revenues, treatment effects, or participant observations. The current completed output is the game specification and experimental design. A later implementation must distinguish protocol fees from experimental point rewards and must not describe the binary fee game as a continuous auction.

F.2 Linked-artifact parity

Artifact	Current status and parity requirement
GitHub	Planned by Lin Zhang; must contain the exact game, parameters, commit, commands, outputs, and limitations.
Hugging Face	Planned by Lin Zhang; must match the treatment design, behavioral hypothesis, and privacy boundary.
A0 poster	Planned by Zijie Wang; must match the model, hypotheses, evidence status, and final links.

F.3 Symposium and evidence status

Team 7 is assigned to Symposium Session A. The symposium has not occurred as of September 26, 2026, so there are no questions, evaluations, discussant comments, or post-symposium revisions. Completed items are the proposal text, payoff matrix, equilibrium benchmark, teaser, treatment design, planned metrics, and division of responsibilities. Planned items are calibration, preregistration, code, experiment, fresh-run analysis, repositories, and A0 poster. Not available are numerical results, participant observations, peer review, and symposium feedback.